

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

TUGATAI AI SAAS PLATFORM

**Omnichannel AI Customer Engagement & Business
Automation Platform**

Prepared For

Tugatai AI SaaS Platform

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, technical, architectural, operational, and integration requirements for the Tugatai AI SaaS Platform.

The purpose of this document is to:

- provide a production-ready project blueprint
 - define clear system behavior
 - guide development and implementation
 - establish technical standards
 - define architectural boundaries
 - support scalable SaaS development
 - serve as a reference for future maintenance and scaling
-

1.2 Background & Problem Statement

Modern businesses face major challenges managing customer communication across multiple channels, such as:

- WhatsApp
- Instagram
- Facebook Messenger
- Email
- Phone communication

Businesses frequently experience the following:

- missed customer inquiries
- delayed responses
- fragmented communication
- lack of centralized CRM
- manual quote generation
- poor booking management
- inconsistent support operations
- poor lead tracking
- lack of workflow automation

Traditional CRM systems are often:

- too complex
- not AI-driven
- not optimized for WhatsApp-first businesses
- disconnected from operational workflows

Businesses also struggle to:

- scale customer support
 - automate lead qualification
 - manage omnichannel conversations
 - automate booking and pricing workflows
 - maintain operational consistency
-

1.3 Proposed Solution

Tugatai AI SaaS Platform is proposed as a scalable omnichannel AI business operating system that combines:

- AI communication
- CRM
- workflow automation
- booking automation
- pricing automation
- omnichannel inbox
- AI knowledge systems
- analytics
- SaaS multi-tenancy

into one unified platform.

The system will:

- centralize all customer communication
- automate repetitive workflows
- assist agents using AI
- automate lead management
- support booking workflows
- support branch-based operations
- provide business analytics
- support scalable SaaS deployment

1.4 Scope

In Scope

The system SHALL include:

- Multi-tenant SaaS architecture
- Branch management system
- Omnichannel inbox
- WhatsApp integration
- Instagram integration
- Facebook Messenger integration
- AI assistant
- CRM system
- Workflow automation
- Pricing engine
- Booking engine
- Campaign management
- Knowledge base system
- PDF/DOCX/CSV uploads
- Stripe subscription system
- Flutter Android app
- Analytics dashboard
- Human handoff system
- AI-assisted workflows
- Role-based access control
- Branch-specific agent management

Out of Scope

The following are NOT included in current MVP:

- Inbound AI voice support
- Enterprise SSO
- Full microservice architecture
- LinkedIn integration
- Telegram integration
- Website live chat
- Enterprise workflow designer
- AI phone support center

- White-label reseller portal
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1.5 Target Users

Primary Users

- SMEs
- Logistics businesses
- Service companies
- WhatsApp-first businesses

Secondary Users

- Sales teams
 - Support teams
 - Operations teams
 - Branch managers
 - Business owners
-

1.6 Definitions, Acronyms & Abbreviations

Term	Meaning
AI	Artificial Intelligence
CRM	Customer Relationship Management
KB	Knowledge Base
RAG	Retrieval-Augmented Generation
SaaS	Software as a Service
RBAC	Role-Based Access Control
API	Application Programming Interface
MVC	Model View Controller
JWT	JSON Web Token

ETA	Estimated Time of Arrival
MVP	Minimum Viable Product
DRY	Don't Repeat Yourself
KISS	Keep It Simple Stupid
YAGNI	You Aren't Gonna Need It

1.7 References

Benchmark Platforms

- WIRAL
- Chatwoot
- Moonline

Technology References

- OpenAI API
 - Stripe API
 - Meta APIs
 - PostgreSQL Documentation
 - Express.js Documentation
 - FastAPI Documentation
-

2. OVERALL DESCRIPTION

2.1 Product Perspective

Tugatai AI is a:

- multi-tenant SaaS platform
- omnichannel communication platform
- AI business operating system

The platform is designed using:

- modular monolithic architecture
- scalability-first architecture
- clean separation of concerns

The system is designed to evolve into a scalable SaaS ecosystem supporting multiple industries.

2.2 Product Positioning

The platform is NOT

- a simple chatbot
- a static FAQ bot
- a basic WhatsApp automation tool

The platform IS

- AI-assisted CRM system
 - omnichannel communication hub
 - workflow automation platform
 - AI-powered booking system
 - operational business platform
-

2.3 Product Functions (High-Level Features)

Core Features

- Omnichannel inbox
- AI assistant
- CRM system
- Campaign management
- Workflow automation
- Pricing engine
- Booking engine
- Analytics
- Knowledge base
- AI training
- Human handoff

- Stripe billing
 - Multi-branch operations
-

3. USER CLASSES & CHARACTERISTICS

3.1 Super Admin

Responsibilities:

- manage platform
 - manage tenants
 - manage subscriptions
 - monitor system health
-

3.2 Tenant Owner

Responsibilities:

- manage business account
 - manage branches
 - manage users
 - manage AI configuration
-

3.3 Branch Manager

Responsibilities:

- manage branch operations
 - monitor agents
 - monitor CRM
 - monitor branch analytics
-

3.4 Agent

Responsibilities:

- respond to customers
 - manage leads
 - update CRM
 - handle bookings
-

4. OPERATING ENVIRONMENT

Frontend Environment

- Flutter Web
 - Flutter Android
-

Backend Environment

- Node.js
 - Express.js
-

AI Environment

- FastAPI
 - OpenAI APIs
-

Database Environment

- PostgreSQL
-

Infrastructure Environment

- Docker
 - Nginx
 - Redis
 - AWS/VPS
-

5. DESIGN & IMPLEMENTATION CONSTRAINTS

Architectural Constraints

- Modular monolithic architecture
 - Future microservice compatibility
-

Coding Constraints

- MVC architecture
 - Repository pattern
 - DRY principle
 - KISS principle
 - YAGNI principle
-

Database Constraints

- PostgreSQL required
 - relational data structure
-

AI Constraints

- AI must NOT directly calculate pricing
 - AI must NOT bypass backend validation
-

6. ASSUMPTIONS & DEPENDENCIES

Assumptions

- OpenAI APIs remain available
 - Meta APIs remain accessible
 - Stripe APIs remain available
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Dependencies

- OpenAI
 - Stripe
 - Meta APIs
 - Redis
 - PostgreSQL
-

7. FUNCTIONAL REQUIREMENTS

FR-1 Multi-Tenant Management

The system SHALL:

- support multiple tenants
 - isolate tenant data
 - allow tenant-specific configurations
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FR-2 Branch Management

The system SHALL:

- support multiple branches under one tenant
 - allow branch-specific agents
 - allow branch-specific inboxes
 - allow branch-specific analytics
-

FR-3 Authentication & Authorization

The system SHALL:

- support JWT authentication
 - support RBAC
 - support session management
-

FR-4 Omnichannel Inbox

The system SHALL:

- centralize all messages
 - support AI + human replies
 - support tags and notes
 - support conversation assignment
-

FR-5 CRM System

The system SHALL:

- auto-create leads
- track lead stages
- support notes
- support activity logs

FR-6 AI Assistant

The system SHALL:

- understand customer intent
 - generate responses
 - retrieve KB information
 - summarize conversations
-

FR-7 Knowledge Base System

The system SHALL:

- upload PDF files
 - upload DOCX files
 - upload CSV files
 - generate embeddings
-

FR-8 Pricing Engine

The system SHALL:

- calculate deterministic pricing
 - support country-specific pricing
 - support surcharge logic
-

FR-9 Booking Engine

The system SHALL:

- create bookings

- schedule pickups
 - generate booking references
-

FR-10 Workflow Automation

The system SHALL:

- support trigger-based workflows
 - support follow-up automation
 - support escalation workflows
-

FR-11 Campaign Management

The system SHALL:

- create campaigns
 - schedule campaigns
 - segment audiences
-

FR-12 Analytics

The system SHALL:

- generate conversation analytics
 - generate conversion analytics
 - generate branch analytics
-

FR-13 Stripe Billing

The system SHALL:

- support subscriptions
 - support billing history
 - support invoices
-

FR-14 Android Application

The system SHALL:

- provide inbox access
 - provide CRM access
 - support notifications
-

8. NON-FUNCTIONAL REQUIREMENTS

Performance

- API response under 500ms
 - real-time inbox synchronization
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Scalability

- modular scalability
 - future microservice extraction capability
-

Security

- JWT authentication
 - RBAC authorization
 - secure credential storage
-

Availability

- 99.9% uptime target
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Maintainability

- clean modular architecture
 - reusable services
 - repository abstraction
-

9. EXTERNAL INTERFACE REQUIREMENTS

User Interfaces

- Flutter Web Dashboard
 - Flutter Android App
-

Hardware Interfaces

- Mobile devices
 - Cloud servers
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Software Interfaces

- OpenAI APIs
 - Stripe APIs
 - Meta APIs
-

Communication Interfaces

- HTTPS
 - WebSockets
 - REST APIs
-

10. SYSTEM ARCHITECTURE

High-Level Architecture

Flutter Web + Flutter Android



Node.js + Express Backend



Business Logic Layer



FastAPI AI Layer



PostgreSQL + Redis

Architectural Style

- Modular Monolith
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Major Modules

Auth Module

Tenant Module

Branch Module

Inbox Module

CRM Module

AI Module

Pricing Module

Booking Module

Workflow Module
Campaign Module
Analytics Module
Billing Module
Notification Module
Knowledge Base Module

11. DATABASE DESIGN

Core Database Tables

tenants
branches
users
roles
permissions
channels
conversations
messages
contacts
leads
crm_notes
bookings
pricing_rules
products
campaigns
subscriptions
payments
knowledge_documents
ai_embeddings
workflow_rules
analytics
audit_logs
notifications

Database Relationships

Tenant Relationships

- One tenant → multiple branches
 - One branch → multiple users
 - One branch → multiple conversations
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CRM Relationships

- One contact → multiple conversations
 - One lead → multiple activities
-

12. API DESIGN

Authentication APIs

POST /api/v1/auth/login
POST /api/v1/auth/register
POST /api/v1/auth/refresh
POST /api/v1/auth/logout

Tenant APIs

GET /api/v1/tenants
POST /api/v1/tenants
PUT /api/v1/tenants/:id
DELETE /api/v1/tenants/:id

Branch APIs

GET /api/v1/branches
POST /api/v1/branches

PUT /api/v1/branches/:id
DELETE /api/v1/branches/:id

CRM APIs

GET /api/v1/crm/leads
POST /api/v1/crm/leads
PUT /api/v1/crm/leads/:id

AI APIs

POST /api/v1/ai/respond
POST /api/v1/ai/summarize
POST /api/v1/ai/embed

Campaign APIs

POST /api/v1/campaigns
GET /api/v1/campaigns
POST /api/v1/campaigns/send

13. SECURITY REQUIREMENTS

Authentication Security

- JWT authentication
 - refresh token rotation
-

Authorization Security

- RBAC

- branch-level access control
-

Infrastructure Security

- HTTPS
 - secure environment variables
 - encrypted secrets
-

API Security

- rate limiting
 - request validation
 - webhook validation
-

14. DEPLOYMENT ARCHITECTURE

Recommended Infrastructure

Nginx Reverse Proxy



Flutter Frontend



Node.js Backend



FastAPI AI Service



PostgreSQL + Redis

Infrastructure Stack

- Docker
- Nginx

- PostgreSQL
 - Redis
 - AWS/VPS
-

15. MOSCOW PRIORITIZATION

MUST HAVE

- Multi-tenant system
 - Branch management
 - CRM
 - AI assistant
 - Omnichannel inbox
 - Stripe billing
 - Campaign system
 - Booking engine
 - Pricing engine
 - Analytics
 - Android app
-

SHOULD HAVE

- AI sentiment analysis
 - advanced CRM automation
 - advanced analytics
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COULD HAVE

- Telegram integration
- LinkedIn integration
- AI predictive analytics

WON'T HAVE

- inbound voice AI
 - enterprise SSO
 - white-label reseller system
-

16. REQUIREMENT TRACEABILITY MATRIX (RTM)

Requirement ID	Description	Module
FR-1	Multi-tenant management	Tenant Module
FR-2	Branch management	Branch Module
FR-3	Authentication	Auth Module
FR-4	Omnichannel inbox	Inbox Module
FR-5	CRM management	CRM Module
FR-6	AI assistant	AI Module
FR-7	Knowledge uploads	KB Module
FR-8	Pricing engine	Pricing Module
FR-9	Booking engine	Booking Module
FR-10	Workflow automation	Workflow Module
FR-11	Campaign management	Campaign Module
FR-12	Analytics	Analytics Module
FR-13	Stripe billing	Billing Module
FR-14	Android app	Mobile Module

17. SUPPORTING INFORMATION

Engineering Principles

- DRY
 - KISS
 - YAGNI
-

Architecture Goals

- maintainability
 - scalability
 - modularity
 - operational efficiency
-

Future Scalability

Modules should support future migration into:

- microservices
 - distributed architecture
 - async processing systems
-

18. FUTURE SCALABILITY STRATEGY

Future Microservice Candidates

- AI service
- workflow engine
- analytics engine
- campaign engine

- notification service
-

Future Features

- inbound voice AI
 - enterprise automation
 - advanced AI orchestration
 - white-label SaaS
-

19. CONCLUSION

Tugatai AI SaaS Platform is designed as a scalable modular omnichannel AI business operating system capable of supporting:

- AI customer engagement
- CRM operations
- workflow automation
- pricing automation
- booking workflows
- campaign management
- branch operations
- multi-tenant SaaS growth

The architecture prioritizes:

- scalability
- maintainability
- modularity
- operational efficiency
- future SaaS commercialization
- enterprise-grade extensibility